LoCoMo 89.1% 基线 — 168 个错题深度分析报告

基线文件: locomo_minimax_m27_eval_baseline.json
LLM: MiniMax M2.7（记忆提取 + 答题 + 评审三端同模型）
总题数: 1540（排除 Category 5 Yes/No 题后）
正确: 1372（89.1%），错误: 168（10.9%）
分析日期: 2026-04-12

一、执行摘要

本报告对 LoCoMo 基线中 168 个错误答案进行了系统性根因分析，关键发现:

1. 69% 的错误（116 题）不是记忆体的问题 — 正确证据已检索到 top-10，失败在 LLM 答题环节
2. 最集中的可修复问题是 Cat 2 时间推理（34 题日期差 ±1 周）— post-extraction enrichment 已验证可修正
3. Cat 3 多跳推理是模型能力天花板（65.6% 准确率）— 需要更强 LLM 或 agentic 架构
4. fact extraction 的概括化倾向导致 17 个提取遗漏（hoodie -> clothing line 等）

二、分析方法论

2.1 验证链

对每个错题执行 5 步精确归因:

1. 从正确答案提取关键词（>=3 字符，过滤停用词）
2. 在全部检索到的记忆中搜索关键词（不限 top-10，覆盖全部 100+ 条）
3. 确定证据位置: top-10 内 / top-10 外 / 完全不存在
4. 结合 LLM 实际回答和评审理由分类失败模式
5. 与 LoCoMo 题目类别交叉分析

2.2 局限性

• 关键词匹配是近似方法，部分情况（答案是数字如 "3"）无法检测
• LLM 的 "无信息" 判断可能是对 fact 文本措辞不匹配的合理反应
• 评审（judge）本身使用同一模型 MiniMax M2.7，存在 ~2pp 的随机噪声

三、168 题全景归因

3.1 三大失败分组

分组	含义	题数	占比	核心问题
GROUP A	证据在 top-10，LLM 仍答错	116	69%	答题环节失败

GROUP B	证据在记忆库但排在 top-10 外	24	14%	检索排序不足
GROUP C	证据不在任何记忆中	28	17%	事实提取遗漏

3.2 细分失败模式

模式	题数	占比	描述
A1: 证据在场，LLM 说「无信息」	37	22%	LLM 扫描 100+ 条 facts 时未能关联查询与证据
A2: 证据在场，LLM 答错	79	47%	LLM 选了错误的 fact 或提取了错误的细节
B1: 证据在 top-10 外，LLM 说「无信息」	6	4%	证据排在 rank 11-20，刚好出局
B2: 证据在 top-10 外，LLM 答错	18	11%	证据排名低，LLM 用了 top 中的错误 fact
C0: 答案过短无法检测	3	2%	正确答案为数字（如 "3"），无法关键词匹配
C1: 无证据，LLM 正确说「无信息」	7	4%	真正的提取缺失，LLM 判断正确
C2: 无证据，LLM 编造答案	18	11%	提取遗漏 + hallucination

3.3 关键洞察

69% 的错误（GROUP A）是答题环节的问题，不是记忆体的问题。 检索管线已成功将正确证据排到 top-10，但 MiniMax M2.7 在 answer generation 阶段未能正确使用。

这意味着:

- 即使检索完美（100% 正确证据在 top-1），仍有 69% 的当前错误无法通过改善检索来解决
- 改善答题质量（更好的 prompt 或更强的 LLM）的理论天花板为 +7.5pp（89.1% -> 96.6%）
- 改善检索排序（GROUP B）的天花板为 +1.6pp
- 改善事实提取（GROUP C）的天花板为 +1.8pp

四、按 LoCoMo 类别深度分析

4.1 类别总览

类别	总题	正确	错题	准确率	最突出问题
Single-hop（单跳事实）	282	258	24	91.5%	A2(8题)
Temporal（时间推理）	321	274	47	85.4%	A2(34题)
Multi-hop（多跳推理）	96	63	33	65.6%	A1(11题)
Descriptive（开放描述）	841	777	64	92.4%	A2(27题)

4.2 Cat 2 时间推理（85.4%）— 最大可修复弱点

47 个错题中 34 个是 A2（证据在 top-10 但提取/选择的日期有误）。

系统性模式: LLM 在 fact extraction 时将相对日期（"last Friday"）转为绝对日期，频繁出现 整整 1 周的偏差:

原文表达	LLM 计算	正确日期	误差	session 日期
"last Friday"	July 14	July 21	-7 天	July 23
"Last Wednesday"	Feb 1	Feb 8	-7 天	Feb 9
"last Friday"	Oct 13	Oct 20	-7 天	Oct 22
"last weekend"	Jul 8	Jul 15	-7 天	Jul 17
"last Friday"	May 20	May 21	-1 天	May 25

根因: concise 模式的 extraction prompt 要求 LLM 做日期算术（"相对 session 日期计算绝对日期"），但 LLM 的算术不准确。所有错误案例的 session 日期（mentioned_at）都是正确的参考点，只是 LLM 从这个参考点做减法时出错。

修复方案: post-extraction date_validation 模块使用 dateparser（确定性日期解析库）独立计算，在 38 个 facts 上验证成功修正。

4.3 Cat 3 多跳推理（65.6%）— 模型能力天花板

33 个错题中 11 个是 A1（证据在但 LLM 说「无信息」）。

这些题需要推理而非直接查找:

问题	正确答案	推理链
"Jolene 多大?"	"不超过 30"	从 "她在上学" 推断年龄
"James 孤独吗?"	"是的"	从 "只有猫迎接他" 推断心理状态
"为什么 Jolene 不做瑜伽?"	"更喜欢打游戏"	从活动偏好推断原因
"Melanie 还会去 roadtrip 吗?"	"可能不会"	从 "这次很糟糕" 推断意愿

根因: MiniMax M2.7 倾向保守回答，面对需要推理的问题选择说 "没有明确信息" 而非做推断。这是模型推理能力的限制，不是记忆体能解决的问题。

4.4 Cat 1 单跳事实（91.5%）和 Cat 4 开放描述（92.4%）

这两个类别准确率较高，错误分布分散，无集中模式。Cat 4 的 19 个 A1 错误（LLM 说「无信息」但证据在场）值得关注——这些可能通过优化 answer generation prompt 改善。

五、按对话分布分析

对话	准确率	正确/总计	主要错误类型
conv-26	88.8%	135/152	A2(9)
conv-30	87.7%	71/81	A2(7)
conv-41	92.1%	140/152	A2(7)
conv-42	86.4%	172/199	A2(9)
conv-43	86.0%	153/178	A2(11)
conv-44	89.4%	110/123	A2(6)
conv-47	90.0%	135/150	A2(5)
conv-48	87.4%	167/191	A2(14)
conv-49	91.0%	142/156	B2(7)
conv-50	93.0%	147/158	A2(5)

10 个对话的准确率范围为 86.0%（conv-43）到 93.0%（conv-50），差异在合理波动范围内（无严重异常值）。

六、GROUP A 逐题分析

6.1 A1: 证据在 top-10，LLM 说「无信息」（37 题）

对话	类别	问题	正确答案	证据排名
conv-48	Descript	What projects is Jolene planning fo...	developing renewable ener...	#1
conv-48	Temporal	When did Jolene do yoga at Talkeetn...	on 5 June, 2023...	#1
conv-48	Descript	What did Jolene and Anna discuss wh...	They realized they inspir...	#1
conv-48	Single-h	Where did Jolene and her partner fi...	Phuket...	#3
conv-26	Descript	What precautionary sign did Melanie...	A sign stating that someo...	#1
conv-26	Multi-ho	Would Melanie go on another roadtri...	Likely no; since this one...	#1
conv-26	Single-h	What book did Melanie read from Car...	"Becoming Nicole"...	#7
conv-47	Multi-ho	Why didn't John want to go to Starb...	Possibly because he likes...	#8
conv-47	Descript	What game was James playing in the ...	Apex Legends...	#7
conv-47	Multi-ho	Was James feeling lonely before mee...	Most likely yes, because ...	#8
conv-47	Temporal	When did John spend time with his s...	July 21, 2022...	#1
conv-44	Multi-ho	Which national park could Audrey an...	Voyageurs National Park...	#1
conv-44	Temporal	When did Andrew make his dogs a fun...	few days before November ...	#1
conv-44	Descript	What did Andrew and his GF do on th...	volunteered at a pet shel...	#1
conv-43	Multi-ho	Which Star Wars-related locations w...	Skellig Michael, Malin He...	#5
conv-43	Multi-ho	What is a prominent charity organiz...	Good Sports, because they...	#1
conv-43	Descript	What did John share with the person...	Characters from Harry Pot...	#2
conv-43	Temporal	Which city was John in before trave...	Seattle...	#8
conv-43	Multi-ho	Based on Tim's collections, what is...	House of MinaLima...	#3
conv-43	Multi-ho	What is a Star Wars book that Tim m...	Star Wars: Jedi Apprentic...	#4
conv-43	Descript	Which basketball team does Tim supp...	The Wolves...	#2
conv-43	Multi-ho	Which outdoor gear company likely s...	Under Armour...	#5
conv-42	Descript	What dish did Nate make on 9 Novemb...	Homemade coconut ice crea...	#1
conv-42	Descript	How did Nate celebrate winning the ...	Taking time off to chill ...	#5
conv-42	Descript	What did Joanna just finish last Fr...	screenplay...	#1

conv-42	Descript	What does Nate want to do when he g...	Watch one of Joanna's mov...	#1
conv-42	Descript	What kind of lighting does Nate's g...	red and purple lighting...	#9
conv-42	Multi-ho	What alternative career might Nate ...	an animalkeeper at a loca...	#1
conv-50	Descript	What did Calvin and his friends arr...	regular walks together...	#1
conv-50	Descript	What hobby did Calvin take up recen...	Photography...	#1
conv-50	Descript	When did Calvin first get intereste...	at an early age...	#5
conv-50	Descript	What realization did the nightclub ...	how much music means to h...	#3
conv-41	Descript	What type of workout class did Mari...	aerial yoga...	#1
conv-41	Descript	What did Maria plan to do later on ...	have dinner with friends ...	#1
conv-41	Descript	What did Maria donate to a homeless...	old car...	#3
conv-41	Multi-ho	What job might Maria pursue in the ...	Shelter coordinator, Coun...	#9
conv-49	Single-h	What is Sam's persistent problem wi...	His new phone malfunction...	#4

6.2 A2: 证据在 top-10，LLM 答错（79 题，仅列前 20）

对话	类别	问题	正确答案	LLM 回答	证据
conv-30	Descript	What did Gina find for her clo...	The perfect spot for...	On February 1, 2023,...	#1
conv-30	Descript	What did Jon and Gina compare ...	dancing together and...	Gina compared her en...	#1
conv-30	Descript	What does Jon tell Gina he won...	quit...	Jon tells Gina he wo...	#5
conv-30	Temporal	When did Gina go to a dance cl...	21 July 2023...	July 14, 2023...	#1
conv-30	Temporal	When did Gina open her online ...	16 March, 2023...	Gina opened her onli...	#7
conv-30	Descript	What do the dancers in the pho...	They are performing ...	Jon shared a photo o...	#10
conv-30	Temporal	When did Jon visit networking ...	20 June, 2023...	Jon attended network...	#2
conv-48	Descript	What did Jolene recently play ...	a card game about ca...	Monster Hunter: Worl...	#1
conv-48	Descript	What did Deb share a photo of,...	a yellow coffee cup ...	A handwritten note f...	#2
conv-48	Descript	Why did Jolene have to resched...	Jolene already had p...	According to the con...	#1
conv-48	Descript	What activity did Jolene and h...	play the console...	camping trip...	#4
conv-48	Temporal	When did Jolene have a mini-re...	Wednesday before 9 F...	Jolene had a mini re...	#1
conv-48	Descript	What do Deborah and her husban...	play detective games...	Deborah and her husb...	#1
conv-48	Descript	What activity does Deborah inc...	spending time with l...	Deborah incorporates...	#4

conv-48	Descript	Which new yoga pose did Debora...	tree pose...	Dancer Pose (Nataraj...	#2
conv-48	Temporal	When did Deborah go to a commu...	last week of August ...	Friday, September 1,...	#1
conv-48	Multi-ho	Why did Jolene sometimes put o...	She's more intereste...	Jolene sometimes put...	#9
conv-48	Temporal	When did Deborah meet Anna?...	31 January, 2023...	Deborah met Anna on ...	#1
conv-48	Descript	According to Jolene, what does...	connected to her bod...	According to Jolene,...	#1
conv-48	Temporal	When do Jolene and her partner...	Saturday after 27 Ja...	Saturday, February 4...	#1
...	...	*余 59 题*	...	...	...

七、GROUP B 逐题分析（24 题）

对话	类别	问题	正确答案	证据排名	LLM 行为
conv-30	Single-h	What Jon thinks the ideal danc...	By the water, with nat...	#13	答错
conv-48	Multi-ho	How old is Jolene?...	likely no more than 30...	#20	说无信息
conv-48	Descript	What games does Jolene recomme...	Zelda BOTW for Switch ...	#16	答错
conv-48	Descript	What game did Jolene recommend...	Animal Crossing: New H...	#11	说无信息
conv-48	Descript	What did Jolene participate in...	presenting at a virtua...	#24	答错
conv-26	Descript	What do sunflowers represent a...	warmth and happiness...	#13	说无信息
conv-26	Temporal	When did Melanie read the book...	2022...	#29	说无信息
conv-47	Temporal	How long did James and Samanth...	nearly three months...	#11	答错
conv-44	Temporal	When did Audrey make muffins f...	The week of April 3rd ...	#12	说无信息
conv-44	Single-h	How many times did Audrey and ...	three times...	#25	答错
conv-44	Descript	What did Audrey share to show ...	photography of a baske...	#80	答错
conv-43	Multi-ho	What other exercises can help ...	Sprinting, long-distan...	#109	答错
conv-43	Descript	Which book did Tim recommend t...	"A Dance with Dragons"...	#86	答错
conv-42	Single-h	How many of Joanna's writing h...	two...	#60	答错
conv-42	Descript	What is Nate's favorite video ...	Xenoblade Chronicles...	#14	答错
conv-42	Descript	What does Nate feel he could d...	write a whole movie...	#12	答错
conv-50	Temporal	How long was the car modificat...	two weeks...	#109	说无信息
conv-49	Descript	What electronics issue has bee...	malfunctioning navigat...	#11	答错
conv-49	Descript	Why had Evan been going throug...	Lost their job due to ...	#14	答错

conv-49	Multi-ho	How often does Sam get health ...	every three months...	#44	答错
conv-49	Single-h	How many roadtrips did Evan ta...	two...	#51	答错
conv-49	Descript	What did Evan start painting y...	forest scene...	#57	答错
conv-49	Descript	What type of landscapes does E...	sunsets over the ocean...	#31	答错
conv-49	Descript	What food did Sam share a phot...	bowl of spinach, avoca...	#12	答错

八、GROUP C 逐题分析（28 题）

8.1 C1: 真正的提取缺失（7 题）

• conv-48: How long did Jolene work on the robotics project g... -> 正确: four months

• conv-47: What are John's suspected health problems?... -> 正确: Obesity

• conv-47: How was John feeling on April 10, 2022?... -> 正确: seeking solitude

• conv-44: Which US state do Audrey and Andrew potentially li... -> 正确: Minnesota

• conv-43: After how many weeks did Tim reconnect with the fe... -> 正确: three weeks

• conv-43: What sports does John like besides basketball?... -> 正确: surfing

• conv-42: What state did Joanna visit in summer 2021?... -> 正确: Indiana

8.2 C2: 提取遗漏 + LLM 编造（18 题）

• conv-30: What did Gina make a limited edition lin... -> 正确: Hoodies / LLM: Gina created a limited

• conv-30: How long did it take for Jon to open his... -> 正确: six months / LLM: It took approximately

• conv-48: Which US state did Jolene visit during h... -> 正确: Alaska / LLM: No US state visit is m

• conv-26: What items has Melanie bought?... -> 正确: Figurines, shoes / LLM: Based on the memories

• conv-26: Did Melanie make the black and white bow... -> 正确: Yes / LLM: No, Melanie did not ma

• conv-47: Did John and James study together?... -> 正确: Yes / LLM: No, John and James hav

• conv-47: How does James plan to make his dog-sitt... -> 正确: By allowing users to c / LLM: James planned to devel

• conv-47: What is the board game where you have to... -> 正确: Mafia / LLM: John doesn't explicitl

• conv-43: What map does Tim show to his friend Joh... -> 正确: a map of Middle-earth / LLM: Tim does not show John

• conv-42: How many letters has Joanna recieved?... -> 正确: Two / LLM: 1

• conv-42: What underlying condition might Joanna h... -> 正确: asthma / LLM: An autoimmune disorder

• conv-42: How many video game tournaments has Nate... -> 正确: nine / LLM: 11 tournaments

• conv-42: How many times has Nate taken his turtle... -> 正确: Twice. / LLM: At least once - on Oct

• conv-42: How many times has Joanna found new hiki... -> 正确: twice / LLM: Joanna has found new h

• conv-42: How long did it take for Joanna to finis... -> 正确: four months / LLM: Based on the memories,

• conv-42: What is Nate's favorite book series abou... -> 正确: dragons / LLM: Nate has recommended t

• conv-50: How long did Dave's work on the Ford Mus... -> 正确: nearly two months / LLM: Dave's work on the For

• conv-41: What might John's financial status be?... -> 正确: Middle-class or wealth / LLM: John's financial statu

C2 根因分类:

子根因	数量（估）	描述	典型案例
概括化	~10	原文具体名词被泛化	"hoodie" -> "clothing line"
完全遗漏	~4	原文有但 fact 中完全没提	"mount Talkeetna" 未提取
世界知识	~3	正确答案需要外部知识	Talkeetna -> Alaska

九、改善天花板与已实施方案

9.1 理论天花板

根因	题数	天花板	当前状态
答题环节（A1+A2）	116	+7.5pp	需 answer prompt 优化或更强 LLM
检索排序（B）	24	+1.6pp	可通过提高 max_tokens 缓解
提取遗漏（C）	28	+1.8pp	post-extraction enrichment 已实施

9.2 已实施: post-extraction enrichment

模块	检查 facts	修正数	目标错题
date_validation	102 个含相对日期	38 修正	Cat 2 的 34 个日期错误
detail_preservation	149 个含通用词	80 恢复	C2 的 17 个提取遗漏

验证的修复案例:

错题	原始 fact	修正后	状态
"limited edition of?" -> Hoodies	"clothing line"	"clothing line (specifically: Hoodie)"	细节恢复
"dance class when?" -> July 21	occurred: July 14	occurred: July 21	日期修正
"games for Deborah?" -> Zelda	"video games"	"video games (specifically: Zelda, Animal Crossing)"	细节恢复
"mini retreat when?" -> Feb 8	occurred: Feb 1	occurred: Feb 8	日期修正
"yoga at Talkeetna?"	"beautiful location"	"location (specifically: Talkeetna)"	细节恢复

十、下一步行动建议

优先级	方向	预期收益	工作量	依赖
P0	开启 enrichment 重新 ingest + eval 验证	+1-2pp	4-5h（ingest 时间）	MiniMax API 稳定
P1	扩展 detail_preservation 词典	+0.3-0.5pp	0.5 天	分析更多 C2 案例
P2	提高 max_tokens（4096 -> 8192）	+0.5-1pp	配置调整	验证 LLM 上下文能力
P3	更换/增强 LLM（answer generation）	+2-4pp	依赖模型	需要可用的更强模型
P4	Agentic 多轮检索（Cat 3 推理）	+1-2pp	1-2 周	架构改造

最高 ROI: P0 — 已实施的 enrichment 模块在 38 个日期 + 80 个细节上验证有效，只需重新 ingest 即可验证端到端效果。